12	**(a)**	**(i)**	e.g. linking a (land) telephone to the (local) exchange	B1	[1]
		(ii)	e.g. connecting an aerial to a television	B1	[1]
		(iii)	e.g. linking a ground station to a satellite	B1	[1]
	(b)	**(i)**	attenuation = $10\lg(P_2 / P_1)$	C1	
			total attenuation = 2.1×40 (= 84 dB)	C1	
			$84 = 10\lg(\{450 \times 10^{-3}\} / P)$		
			$P = 1.8 \times 10^{-9}$ W	A1	[3]
			(*answer 1.1×10^{8} W scores 1 mark only*)		
		(ii)	maximum attenuation = $10\lg(\{450 \times 10^{-3}\} / \{7.2 \times 10^{-11}\})$		
			= 98 dB	C1	
			maximum length = 98/2.1		
			= 47 km	A1	[2]